

# Introduction to Mathematical Reasoning & Pattern Spotting!

*Fun festive math puzzles made easy for Grade 2 kids.*



**Can you help Kabir and Diya solve the mystery code?**

**Hello Little Math Genius!** Welcome to the puzzle team. Today, we will learn how to look closely at shapes, colors, and numbers to find hidden rules. You have super math powers—let's use them!

## Welcome to the Diwali Mela!

Our friends **Kabir** and **Diya** are walking around the beautiful, bright Diwali festival ground. Suddenly, they find a very old, magical puzzle box sitting on a table!



### The Secret Locked Box:

The magical box has a big iron lock on it. It does not use a key! Instead, it is locked by a secret pattern code. All around the children are colorful toy stalls, blooming flower decorations, and twinkling lights.

### Your Mission today:

Kabir and Diya need your help to crack the pattern codes! If you follow the steps in this book, you will find the hidden answers, unlock the magic box, and discover the surprise inside!

## What is a Pattern?

A pattern is when things like shapes, colors, or numbers repeat over and over again by following a strict, smart rule!



### The Door Hanging decoration (Toran)

Look at the festive green and orange garland hanging on the stall door:

**Mango Leaf → Marigold Flower → Mango Leaf → Marigold Flower → ?**

What comes next? Yes! A **Mango Leaf** comes next because the rule says they must take turns one by one.

### When you see a puzzle, always ask yourself:

- ★ **What do I see?** Look at all the items carefully.
- ★ **What changes?** Check if things get bigger, spin around, or change colors.
- ★ **What stays the same?** Find what does not change at all.

## More Patterns Around Us!

Patterns are not just for shapes and decorations. Numbers love to make patterns too! You can find number patterns when you play outside or use pocket money.

### 1. The Hopscotch Game

Look at the numbered squares chalked on the ground: **3, 6, 9, 12...**

Can you guess the rule? We are skip-counting by 3s! We hop over 3 numbers every single time. The next square will be **15**!

### 2. Festive Money Notes

Even rupee notes follow rules so people don't get confused:

- **5 Rupee notes:** 5, 10, 15, **20**... (Counting up by 5s)
- **10 Rupee notes:** 10, 20, 30, **40**... (Counting up by 10s)

## Puzzle Walkthrough #1: Spinning Shapes

Let's look at a real Olympiad puzzle together to see how shapes can spin around like a toy wheel. This is called **Visual Rotation**.

### The Triangle Pattern Clue:

Imagine four triangles lined up in a row:

- First triangle points **UP** to the sky.
- Second triangle points **RIGHT** to the side stall.
- Third triangle points **DOWN** to the ground.
- Fourth triangle points **LEFT** to the entry gate.

#### How to solve it:

The shape changes its direction by turning like a clock hand! It goes: *Up* → *Right* → *Down* → *Left*. After it points Left, the whole rule starts over again. So, the next triangle must point **UP**!

## Puzzle Walkthrough #2: The Number Wheel

Now let's try a tricky math number puzzle that looks like a wheel. To solve it, we just need to use our 4 superpower steps!

### Step-by-Step Solving Blueprint:

**Step 1: What do I see?** A round wheel split into sections. Each section has a number like 2, 4, 6, 8, 10...

**Step 2: What is the Rule?** Look at the path of numbers. Every step we take, the number gets bigger by adding 2! ( $2 + 2 = 4$ , then  $4 + 2 = 6$ , then  $6 + 2 = 8$ ).

**Step 3: Find the Missing Piece:** If the last number is 12, we follow our rule and add 2 more.  $12 + 2 = 14$ !

**Step 4: Check Your Work:** Does 14 fit perfectly between 12 and 16? Yes, it does! The mystery is solved!

## The Challenge Zone: Question 1

Are you ready to use your pattern superpowers on your own? Let's crack the first riddle to unlock the magic box's outer layer!



### Question 1: The Rangoli Boundary Pattern

Diya sees a border pattern drawn with colored chalk on the ground. It goes:

**Triangle UP → Triangle RIGHT → Triangle DOWN → ?**

What direction must the next triangle point to keep the sequence correct?

A) Triangle UP

B) Triangle LEFT

C) Triangle RIGHT

D) Triangle DOWN



**Hint:** Look closely at how the triangle turns! What direction comes right after DOWN when a wheel turns around?

## The Challenge Zone: Question 2

Fantastic job! Now let's try a number challenge. Kabir wants to skip across the special squares near the sweets shop.

### Question 2: Hopscotch Skip Counting!

Look at the lucky numbers painted on these stepping stones:

**4 → 8 → 12 → ?**

Which number should Kabir land on next?

A) 14

B) 15

C) 16

D) 18

 **Hint:** What number are we adding to get from 4 to 8? Use your fingers to count! Add that same value to 12 to find the secret answer.



## The Challenge Zone: Question 3

This is a balancing scale puzzle called a **Tarazu Puzzle**. Kabir is visiting a fruit seller who uses clues to weigh sweet fruits.



### Question 3 (Intermediate Level): Balance Clues

Read the two secret math clues on the fruit chart carefully:

- **Clue 1:** 2 Apples weigh exactly the same as 1 Mango.
- **Clue 2:** 2 Mangoes weigh exactly the same as 1 Papaya.



**Question: How many Apples do you need to balance 1 big Papaya?**

A) 2 Apples

B) 4 Apples ☒

C) 6 Apples

D) 8 Apples



**Hint:** If 1 Papaya needs 2 Mangoes, and each of those Mangoes needs 2 Apples, you can add  $2 + 2$  to get the total number of apples!

## Watch Out for Tricks & Review!

Even the smartest math heroes make little mistakes sometimes! Let's learn from our funny friend Mojo so we can avoid tricks:

### ✖ Mojo's Common Pitfalls:

- **Missing the Outers:** Mojo forgot to count the big outer triangle shape when looking at a paper kite decoration! Always check the borders.
- **Rushing the Count:** Mojo counted too fast and skipped a number in a shape row. Slow down and point your pencil at each object one by one!



**Super Math Powers Unlocked!**

**You helped Kabir and Diya solve the secret codes!**  
**The magical box popped open with bright confetti. Keep exploring your world!**